CSCC63 – Week 9 Tutorials

Consider the language BETWEENNESS:

> **INPUT:** A set A of size n, a set T of triples $(a, b, c) \in A^3$, where a, b, and c are distinct.
>
> **QUESTION:** Is there a map $f : A \mapsto \{1, 2, \ldots, n\}$ such that for every $(a, b, c) \in T$, either $f(a) < f(b) < f(c)$ or $f(c) < f(b) < f(a)$?

This langauge shows up in genetic analysis, we won't go into the details of that here.

To see what we mean by this languge definition, suppose we have a set $A = \{a, b, c, d\}$.

- Suppose we also have the set T of triples $\{(a, b, c), (a, b, d), (c, d, a)\}$. This would give us a *yes*-instance of BETWEENNESS, since we can order the four elements as

  $$a, b, d, c.$$

  Note that the ordering of the first triple is maintained, since

  $$a, b, \sqcup, c$$

  is the ordering of the triple (a, b, c). We're allowed to insert the d into the middle of the triple. Similarly, the ordering of the triple (a, b, d) is maintained.

  The ordering of the triple (c, d, a) is reversed, but we also allow that. What we don't allow is a cycling of the triples – so the ordering

  $$a, c, b, d$$

  would violate the order of the triple (a, b, c).

- Suppose instead that we have the set T of triples $\{(a, b, c), (a, b, d), (c, a, d)\}$. If you look through the orderings you should find that none of the ways of ordering these elements follows the order given in the triples (or their reverses). The first two triples constrain c and d to be on the same side of a, and the third triple sets them on the opposite sides. So this would be a *no*-instance.

So that's what the language is asking for.

We'll asume that we know this language is in $\mathcal{NP}$, so we'll be concerned with building a reduction from 3SAT to show that it's $\mathcal{NP}$-complete.

So let's assume we're given an input 3CNF $\langle \phi \rangle$. Using the helpful 3SAT lemma from the Polytime Reductions hndout we'll assume wlog. that the cluses of this ϕ each have three distinct variables.

1. In our 3SAT $\leqslant_p$ 3COL reduction we started with a palette widget: we needed a widget whose purpose was to provide a framework we could use to constrain the rest of the colour choices. We'll want a slightly different framework here, but one that serves the ame purpose.

What I want to do is introduce two characters $\perp$ and $\top$ into A that'll act as brackets for the rest of the characters. We'll later add more characters to A, but as we do we'll want to ensure that the only possible orderings we allow are of the form

$$\perp, \langle \text{everything else} \rangle, \top$$

or

$$\top, \langle \text{everything else} \rangle, \perp$$

.

If we have $A = \{\perp, \top, x_1, x_2, \ldots, x_n\}$, what sort of triples would we add to T to enure this?

2. We'll now try to extend our BETWEENNESS instance to encode truth assignments.

   We'll add a single new element m to our instance to represent a sort of mid-point. For each variable x_i of ϕ we'll also add two new elements x_i and y_i.

   Show how you can use these elements to encode a truth assignment for ϕ.

3. Now, suppose that we relate the elements y_i to the literls $\neg x_i$ (and the elements x_i to the literals x_i). If you place a new element between two literals that are false in given truth assignment, where does it hve to be in your ordering (in terms of your true/false assignments)? What if both literals are true, or if one is true and one false?

   Use your observations here to encode a new BETWEENNESS element c_j for every clause C_j in ϕ, along with a series of triples that ensure that in any truth assignment, c_j can be made true according to your encoding iff the cluase C_j is satisfied.